

Mid-Term Report: Sensor-Adaptive Agricultural Vision Foundation Models

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Abstract

We study continual self-supervised pretraining of official Vision Transformer (ViT) checkpoints on a heterogeneous agricultural image corpus. The project targets transfer across crops, diseases, acquisition conditions, and sensing modalities while minimizing changes to the pretrained backbone. Since the proposal, we have implemented a unified framework supporting ImageNet-, MAE-, DINOv2-, and DINOv3-initialized ViT-S/B/L models; masked image modeling (MIM); DINO-style student-teacher training; and a learnable 1×1 spectral adapter for RGB and multiband inputs. The data layer now streams folders, ZIP and nested-ZIP archives, GeoTIFF, NumPy arrays, and WebDataset shards. A validated catalog contains 5,175,016 RGB images, substantially exceeding the 412,454-image corpus available at proposal time. Engineering evaluation shows successful end-to-end MIM and DINO runs, five-band input support, MIM-to-DINO checkpoint transfer, deterministic resume, and 124 passing automated tests. These results establish implementation readiness, but they are not yet evidence of improved representation quality. Real multispectral coverage, duplicate control, source-balanced sampling, and downstream held-out-domain comparisons remain in progress. The next phase will establish ImageNet baselines and measure linear-probe, few-shot, and fine-tuning performance before scaling pretraining.

1. Introduction

Agricultural vision systems encounter variation that is less prominent in many standard image benchmarks: the same crop changes with growth stage, stress, season, geography, management practice, camera, and illumination. Disease datasets are often composed of centered leaf photographs, whereas deployment may involve cluttered field scenes, unmanned aerial vehicle imagery, or multispectral rasters. Labels for these domains are costly and fragmented among species identification, disease classification, weed recognition, detection, segmentation, counting, and temporal monitoring. Consequently, a model that performs well on a random split of one dataset may transfer poorly to a new farm or sensor.

Our project asks whether continual self-supervised pretraining can adapt an official pretrained ViT to agricultural visual structure without discarding the benefits of generic pretraining. We compare reconstruction-based MIM with view-invariant DINO-style self-distillation and will test a sequential MIM→DINO schedule. A lightweight spectral adapter maps an arbitrary number of aligned bands to the three-channel interface of an official RGB ViT. This design keeps the backbone unchanged and makes sensor adaptation directly ablatable.

At mid-term, the main achievement is a research-ready implementation and a much larger indexed corpus. The software now supports both objectives, multiple official checkpoint families, archive-scale streaming, reproducible state, and downstream task runners. However, the scientific hypothesis remains open: smoke-test losses and passing unit tests demonstrate correctness, not superior transfer. We therefore report engineering results separately from scientific results and reserve performance claims for controlled downstream experiments.

2. Background and Related Work

Vision Transformers. ViT represents an image as a sequence of patch tokens and applies a Transformer encoder, demonstrating that large-scale image pretraining can rival convolutional networks [1]. Our system uses official ViT-S, ViT-B, and ViT-L implementations rather than a project-specific transformer, allowing initialization and scale to be controlled across experiments.

Masked image modeling. MAE masks a large fraction of image patches and learns to reconstruct their pixels, producing transferable representations with an asymmetric encoder-decoder design [2]. Related masked-image methods show that reconstruction or token prediction can provide an effective self-supervised signal [3]. Agricultural lesions, venation, leaf edges, and canopy texture motivate testing whether this local reconstruction signal benefits domain adaptation.

Self-distillation. DINO trains a student to match the centered and sharpened predictions of an exponential-moving-average (EMA) teacher under different augmented views [4]. DINOv2 demonstrates that careful data curation and self-distillation can produce robust visual features without labels [5]. Reconstruction and self-distillation emphasize dif-

Table 1. Corpus status at proposal and mid-term. Counts describe indexed files, not unique biological observations.

Property	Proposal	Mid-term
Validated images	412,454	5,175,016
Local storage	35.86 GB	~1 TB
Primary modality	RGB	RGB
Real multispectral archive	No	No
Archive streaming	Audit only	ZIP/TAR/WDS

ferent invariances, motivating MIM-only, DINO-only, and sequential training under a shared data and backbone stack.

Agricultural transfer and spectral adaptation.

PlantVillage helped establish deep learning for image-based plant-disease recognition, but its controlled backgrounds also illustrate the difference between benchmark and field conditions [6, 7]. Remote-sensing foundation models such as SatMAE adapt masked pretraining to temporal and multispectral satellite imagery [8]. In contrast, our adapter preserves an official RGB backbone and learns a small sensor-specific projection, which enables a controlled comparison against RGB-only input and direct patch-embedding expansion.

3. Approach

3.1. Data and Sampling

The proposal reported 412,454 images from 13 downloaded datasets. The current validated catalog indexes 5,175,016 loader-visible images across approximately 1 TB of ZIP and TAR archives. Sources cover plant species, diseases, seedlings, weeds, and crop imagery, but the present catalog remains overwhelmingly RGB. Each catalog record stores a stable sample key, source identity, and portable archive-member path. Large sources are converted to roughly 1 GB WebDataset shards after RGB conversion, bounded resizing, and validation. Parallel workers process one archive at a time to reduce archive-open and seek overhead.

To prevent a large species collection from dominating training, the planned sampler first selects a source and then samples within that source. Validation splits are group-disjoint when source metadata permits. Before final experiments, we will freeze the manifest, record licenses and versions, and remove exact and near-duplicate clusters before creating train and held-out-domain partitions.

3.2. Sensor-Adaptive Backbone

For an input $\mathbf{x} \in \mathbb{R}^{C \times H \times W}$, a learnable 1×1 convolution projects the C sensor bands into the three channels expected by an official pretrained ViT:

$$\mathbf{z}_{h,w} = A\mathbf{x}_{h,w} + \mathbf{b}, \quad A \in \mathbb{R}^{3 \times C}. \quad (1)$$

For RGB, A is initialized to the 3×3 identity and $\mathbf{b} = 0$. For multiband input, available RGB-like channels initialize the corresponding outputs; weights for additional bands begin at zero but remain trainable. The adapted image is normalized with the statistics associated with the selected official checkpoint, then patchified by the unchanged ViT. The adapter adds only $3C + 3$ parameters and cleanly separates sensor mixing from spatial feature learning.

3.3. Self-Supervised Objectives

In MIM, a random set $\mathcal{M}$ masks 75% of patch tokens. A linear reconstruction head predicts the adapted RGB patch $\mathbf{p}_i$ from encoded token $\mathbf{h}_i$; loss is evaluated only at masked positions:

$$\mathcal{L}_{\text{MIM}} = \frac{1}{|\mathcal{M}|} \sum_{i \in \mathcal{M}} \|g(\mathbf{h}_i) - \mathbf{p}_i\|_2^2. \quad (2)$$

For DINO-style training, the student receives global and local crops while the teacher receives paired global crops. With student temperature τ_s , teacher temperature τ_t , and running center $\mathbf{c}$, the cross-view loss is

$$\mathcal{L}_{\text{DINO}} = - \sum_k q_t^{(k)} \log q_s^{(k)}, \quad q_t = \text{softmax}((\mathbf{o}_t - \mathbf{c})/\tau_t). \quad (3)$$

The teacher parameters, including adapter, backbone, and projection head, are updated by $\theta_t \leftarrow m\theta_t + (1 - m)\theta_s$. The framework supports ImageNet, MAE, DINOv2, and DINOv3 initialization and an explicit MIM→DINO handoff that starts a new optimization run from a compatible MIM checkpoint.

3.4. Training and Evaluation Protocol

Training uses BF16 mixed precision, gradient accumulation, optional gradient checkpointing, cosine scheduling, and multi-process data loading. Every run stores resolved configuration, environment and Git metadata, metrics, figures, best/latest checkpoints, optimizer state, and random-number-generator state. WebDataset provides CPU streaming; NVIDIA DALI is available on Linux for GPU JPEG decoding. Development began on a single RTX 4090, and the next stage will use Auburn’s Slurm cluster for longer controlled runs.

Scientific evaluation will compare the unadapted official checkpoint, RGB-only continual pretraining, MIM, DINO, sequential MIM→DINO, frozen versus trainable adapters, and direct patch-embedding expansion. Classification will report accuracy and macro-F1; detection, segmentation, and counting will use mAP, IoU/Dice, and MAE/RMSE as appropriate. Linear probing, few-shot subsets, and full fine-tuning will be repeated across at least three seeds. The primary test sets will hold out whole sources, crops, geographies, seasons, or sensors, with mean, standard deviation, and paired comparisons reported where splits are shared.

Table 2. Mid-term implementation validation. “Pass” indicates successful execution and expected artifacts, not a downstream accuracy result.

Check	Input	Result
MIM, official ImageNet ViT-S	5-band NPY	Pass
DINO, official ImageNet ViT-S	RGB PNG	Pass
DINO, official DINOv2 ViT-S	RGB PNG	Pass
MIM→DINO initialization	5-band NPY	Pass
Two-epoch checkpoint resume	RGB/5-band	Pass
Automated test suite	124 tests	124 pass
Shard builder smoke test	ZIP/plain	2 pass

4. Intermediate and Preliminary Results

4.1. Verified Engineering Results

Table 2 summarizes completed checks. End-to-end runs produced checkpoints, metrics, training curves, MIM reconstructions, DINO augmented views, and student–teacher similarity matrices. Five-band float32 arrays passed through the learnable adapter and official ImageNet ViT-S. DINO also ran from an official DINOv2 ViT-S checkpoint with its patch-14 geometry, and a compatible MIM checkpoint successfully initialized a new five-band DINO run. Two-epoch resume tests restored optimizer, scheduler, scaler, history, RNG, and loader state.

The automated suite covers RGB, NPY, GeoTIFF, ZIP and nested-ZIP loading; group-disjoint splits; NoData handling; undersized-image padding; official model selection; adapter initialization; MIM and DINO tensor behavior; EMA updates; paired crop replay; strict configuration parsing; atomic checkpointing; and RoPE-based DINOv3 backbones. Recent shard-builder tests additionally verify parallel archive grouping and valid TAR output. These checks reduce the risk that a long pretraining run fails because of data format, checkpoint, or model geometry errors.

4.2. Interpretation and Remaining Gaps

The corpus expansion and successful pipelines satisfy the mid-term engineering milestone: the planned model can now be trained reproducibly at archive scale. The results also support the feasibility of the spectral interface because the same official backbone accepts both RGB and five-band tensors without replacing its patch embedding. They do *not*, however, test whether additional bands improve predictions; the five-band data used for verification are synthetic, and the catalog lacks a verified real multispectral archive.

No model is yet shown to outperform its official ImageNet or DINO initialization on a held-out agricultural benchmark. We therefore make no claim of improved accuracy, robust-

ness, or label efficiency at this stage. The largest immediate risks are duplicate leakage, imbalance from the 306k-image PlantNet subset and other large sources, RGB-only bias, and I/O saturation during multi-GPU training. The first three are addressed by manifest freezing, duplicate clustering, source-balanced sampling, and external domain splits; the last will be measured by comparing CPU WebDataset and DALI throughput before requesting more GPUs.

The next milestone is a ViT-S experiment matrix on frozen data: official initialization, MIM, DINO, and sequential MIM→DINO, followed by linear probing and low-label fine-tuning on source-disjoint disease and species benchmarks. Only recipes that improve repeated-seed transfer will scale to ViT-B. In parallel, we will add a licensed real multi-spectral source and evaluate frozen/trainable adapters against direct patch-embedding expansion. This order prioritizes evidence over model size and keeps the final conclusions aligned with the data actually available.

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